Saketh Ayyagari

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Education

University of Maryland, College Park College Park, MD
Bachelor of Science: Computer Science, Minor in Robotics and Autonomous Systems *May 2028*

Experience

Machine Learning Engineer Feb 2026 - Present
Children's National Hospital *College Park, MD*

- Contracted through UMD's App Development Club to develop a mobile application to detect safe vs. unsafe football tackles using pose estimation and computer vision algorithms to support patient safety.
- Designed and implemented recording and clip analysis features using Figma while collaborating with 5 full-stack engineers
- Collected and annotated 20+ videos of football plays using Roboflow to ensure an accurate split of testing and training data.

Computer Vision and Navigation Software Developer Sep. 2025 - Present
Robotics at Maryland *College Park, MD*

- Collaborating with mechanical, electrical, and software subteams to develop a Mars Rover for Canadian International Rover Challenge.
- Integrated depth camera into a Gazebo simulation to obtain a PointCloud stream at 60 Hz.
- Implemented ROS2 packages for local path planning and autonomous navigation capabilities.

File Systems Intern Sep. 2024 - Jan. 2025
Commvault *Tinton Falls, NJ*

- Developed Python script to visualize file metadata changes across 100+ files over a 7-day interval.
- Implemented anomaly detection functionalities using existing C++ code, file metadata, and SQL queries.
- Implemented false positive detection methods to improve script reliability and accuracy.

Research Intern Jul. 2024 - Aug. 2024
MIT Beaver Works Summer Institute *Cambridge, MA*

- Selected as 1 of 30 students for the Autonomous RACECAR Project.
- Developed computer vision and control algorithms with ROS2 to navigate a scale-model car; placed 3rd out of 7 teams in Grand Prix.
- Published research paper on IMU-based tire blowout detection; accepted to 2025 IEEE Vehicular Technological Conference.

Projects

Simultaneous Localization and Mapping System Feb. 2025 - Jun. 2025

- Implemented SLAM algorithm using 'slam-toolbox' ROS2 package.
- Integrated encoders, IMU, 2D LiDAR and Raspberry Pi 4B into a physical differential drive robot.
- Developed custom ROS2 package in Python/C++ on Ubuntu for teleoperation and motor/sensor integration.

Predicting Movement Direction with Machine Learning Mar 2024 - Jan 2025

- Trained SVMs and Random Forests using scikit-learn, and Neural Networks with Tensorflow on neural spike data to predict the direction of a mammal's movement.
- Tested accuracy and loss of four models on a dataset of 143 neurons with spike data over 158 trials.
- Drafted a 7-page research paper outlining methodology and results.

Neural Network for Pneumonia Detection Jul 2023

- Built convolutional neural networks with Tensorflow for detecting pneumonia with 83% peak accuracy.
- Augmented dataset of 100+ images of chest X-Rays with rotations, scalings, and other transformations with OpenCV.
- Applied transfer learning techniques by implementing VGG16 pretrained model to improve accuracy
- Deployed model to Python-based web application with Streamlit for users to test.

Technical Skills

Languages: Python, C++, Java, SQL, C
Frameworks/Libraries: Tensorflow, PyTorch, NumPy, Scikit-learn, ROS2, Gazebo, OpenCV, Streamlit.
Tools: Git, GitHub, Docker, VS Code, Visual Studio, PyCharm, Eclipse, Roboflow, Microsoft Office, Google Suite